Protecting RAG Systems from Retrieval Spoofing Attacks

Week 9 — Interim Presentation

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Project Overview

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Motivation

Retrieval-Augmented Generation (RAG) systems retrieve external documents before generating answers with an LLM. These systems are widely deployed in search copilots, medical forums, and knowledge bases.

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Problem

Attackers can inject synthetic "spoof" chunks into the retrieval corpus. These chunks appear semantically relevant, rank highly during retrieval, and contain misleading information—causing the LLM to generate incorrect answers.

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Goal

Develop a retrieval-stage defense that detects suspicious chunks, re-ranks results before generation, and improves robustness without modifying the LLM architecture.

Updates Since Proposal

Hybrid Retrieval

Added hybrid dense+sparse retrieval to compare robustness across retrieval methods.

Semantic Attraction Attacks

Spoof chunks are optimized to rank highly in dense retrieval using embedding similarity rather than simple keyword overlap.

Retrieval-Stage Defense

Introduced a lightweight retrieval-stage reranking defense using lexical and structural signals.

LLM Judge

Added GPT-4o-mini evaluation for answer correctness and misleading retrieval behavior.

Novelty: Our attacks target dense retrieval embeddings directly, creating semantically attractive spoof chunks that manipulate retrieval rankings before answer generation.

Previous Work

Paper / Framework	Task & Year	Main Idea	Relation to Our Work
PoisonedRAG PoisonedRAG GitHub	Retrieval poisoning 2025	Inject misleading retrieval documents	Similar attack scenario
BERT-Attack BERT-Attack GitHub	Adversarial NLP 2020	Embedding-preserving adversarial text	Similar semantic attack principles
TextAttack TextAttack GitHub	NLP attack framework 2020	Generate adversarial textual examples	Inspired spoof generation
OpenAttack OpenAttack GitHub	Adversarial text attacks 2021	Benchmark adversarial NLP robustness	Related evaluation methodology

Key Insight

Most previous work focuses on prompt injection, jailbreaks, and input manipulation. Our project addresses retrieval-stage semantic spoofing, corpus poisoning attacks, and retrieval robustness before LLM generation.

•1,500 spoof chunks
•300 attacked queries

Dataset & Semantic Spoof Generation

Dataset

SQuAD v1.1 corpus
5,000 source documents
1,000 validation questions available

Chunking & Retrieval

120-word chunks with 30-word overlap
Dense retrieval: BGE-M3 embeddings + FAISS
Sparse retrieval: BM25Okapi
Hybrid retrieval: Weighted dense–sparse fusion

Semantic Spoof Generation

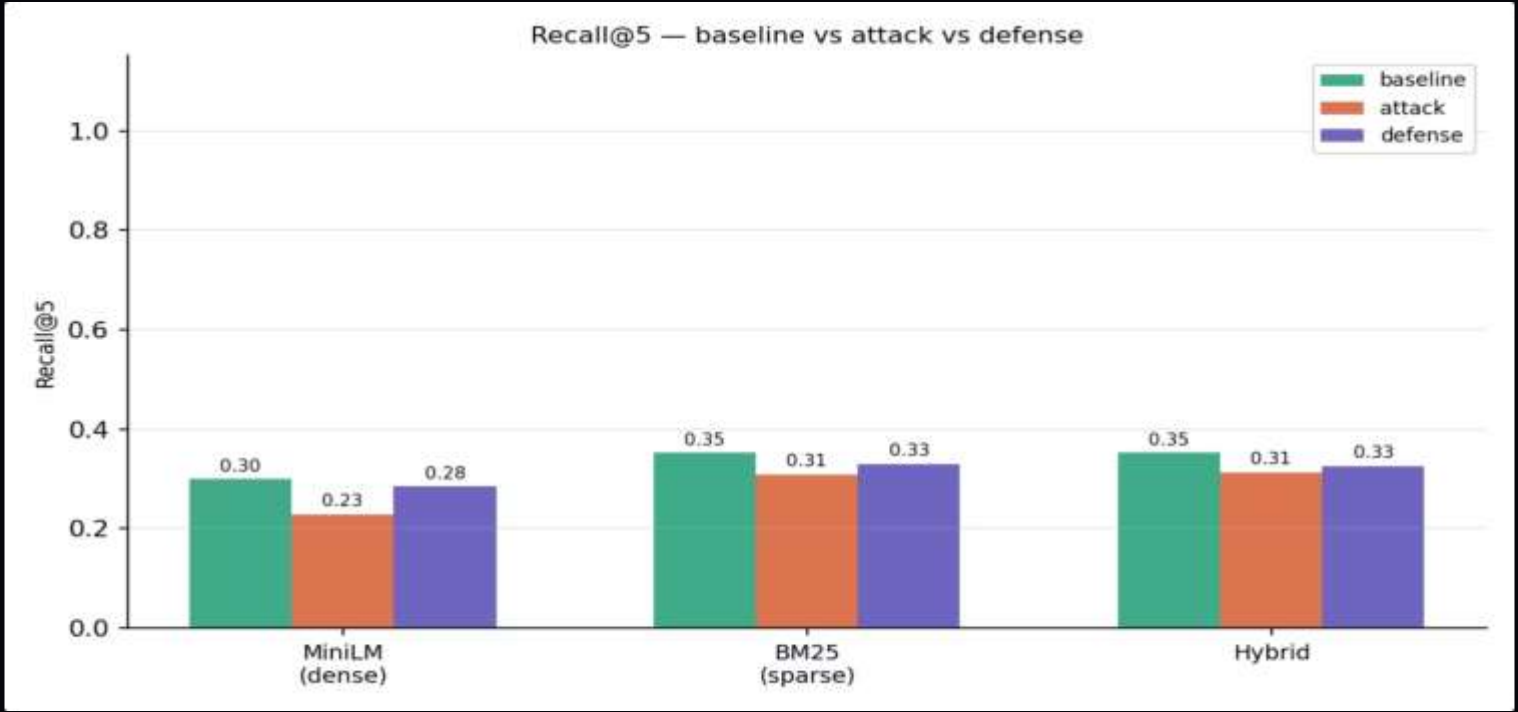
GPT-4o-mini generated spoof chunks
5 attack styles: encyclopedic, news, textbook, forum, abstract
Semantically relevant but evidence-free content
Optimized to attract dense retrievers

Attack Distribution & EDA

5 balanced spoof generation styles.
~1,500 spoof chunks generated.
300 attacked queries.
Dense retrieval shows higher spoof attraction than BM25.

Dense retrieval is more vulnerable to semantically attractive spoof chunks than sparse retrieval methods.

Baseline Retrieval & Attack Results



Key Findings: Dense retrieval is highly vulnerable to semantic spoofing. BM25 shows greater robustness. Hybrid retrieval achieves strongest baseline. Defense partially recovers Recall@5 without labeled training data.

Progress & Next Steps

✓ Dataset Preparation SQuAD v1.1 loading and validation	✓ Chunking Pipeline Document splitting and indexing
✓ Retrieval Systems Dense, sparse, and hybrid methods	✓ Spoof Generation 5 styles × 300 queries = 1,500 chunks
✓ Defense Layer Retrieval-stage reranking defense	✓ Evaluation Recall@5 and LLM Judge metrics

Future Work

1. **Improve Baseline Retrieval**
   Increase clean Recall@5 before applying defense mechanisms.
2. **Stronger Semantic Attacks**
   Explore embedding-optimized retrieval poisoning strategies.
3. **Stronger Defenses**
   Explore more robust reranking and filtering strategies.
4. **Finalize Results**
   Complete evaluation, analysis, and presentation of findings.

> Final Goal: Evaluate the vulnerability of dense, sparse, and hybrid retrieval systems to semantic retrieval poisoning and measure defense effectiveness.